



Special assignment FPGA-Based System Design

Vending Machine Controller - Using Verilog HDL

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INDEX:

1. Objective, abstract and keywords

1.1 Objective and Abstract

1.2 Keywords

2. Introduction

3. Literature survey

4. Limitations & Disadvantages

5. Proposed solution

6. State Diagram

7. Algorithm

8. Flow chart of the code

9. RTL and TTL view

10. Simulation and results

11. Conclusion

12. References

13. Appendix

13.1 Verilog code for the implemented system

1. Objective and Abstract:

1.1. Objective and Abstract:

In this report, a design and implementation of an FPGA-based vending machine controller system will be presented. The system will be implemented based on the Verilog hardware description language. Additionally, the design will be based on a Finite State Machine (FSM) to control the vending process. The system will be able to receive coins with varying denominations and will be able to check if the received amount is enough to allow the product to be dispensed. The system will be able to return change if the received amount is higher than the required amount.

The proposed model will be simulated using ModelSim Altera. Additionally, the design will be synthesized to check the implementation based on the RTL and Technology Level. The system will be implemented on an FPGA development board, with toggle switches used for input to represent coins inserted into the system. Similarly, the output will be represented by LEDs.

1.2. Keywords:

FPGA, Verilog HDL, Vending Machine Controller, Finite State Machine (FSM), ModelSim Altera, Quartus, RTL Design, Digital System Design, FPGA Implementation, Simulation, Synthesis, Hardware Description Language, Switch Inputs, LED Outputs, Automation Systems, Digital Controller

2. Introduction :

A vending machine is an automated device used to dispense the desired product after receiving the required money from the user, it is widely used in public places such as airport, railway station, in canteen, offices etc. since it provides an easy and convenient method of purchasing items without the intervention of human. Modern vending machine uses online method of method like UPI for receiving payment which it processes and then dispenses the product.

3. Literature survey :

- Vending machines are automated machines used for the dispensing of products after receiving the requisite amount of money from the customer.
- Vending machines reduce the need for human intervention and are deployed in public places such as offices, airports, railway stations, and educational institutions.
- Vending machines use digital controllers for carrying out various operations such as coin detection, balance calculation, product vending, and returning change to the customer.
- Automation improves the efficiency, accuracy, and speed of the vending machines.
- FPGA-based control systems are highly reliable and flexible for implementing vending

machine controls.

4. Limitations & disadvantages:

In traditional vending machine systems, control operations were often implemented using microcontroller based hardware. While microcontrollers are widely used in automation systems, they have several limitations when handling complex digital control tasks such as vending machine operations.

1. **Limited Processing Capability:**
Microcontrollers generally have limited processing power compared to FPGA based systems. This restricts the ability to implement more complex control logic and high speed operations in vending machine systems.
2. **Limited Input/Output Pins:**
Microcontrollers have a fixed number of I/O pins which limits the number of inputs and outputs that can be connected directly. This may require additional hardware when multiple sensors, buttons, or display devices are needed.
3. **Timing Constraints:**
In some cases, microcontrollers may face difficulty meeting strict timing requirements, especially when handling multiple operations such as coin detection, product selection, and change return simultaneously.
4. **Sequential Execution:**
Microcontrollers execute instructions sequentially, which can slow down system performance when multiple operations must be processed at the same time.
5. **Memory Limitations:**
Microcontrollers have limited memory resources which can restrict the implementation of complex algorithms or additional features in vending machine systems.
6. **Scalability Issues:**
Expanding the system to support more products, payment methods, or advanced control features may become difficult with microcontroller based designs.
7. **Hardware Flexibility:**
Microcontroller based systems often require hardware modification when design changes are needed, while FPGA based systems allow easier modification through hardware description language programming.

Due to these limitations, FPGA based vending machine controllers provide a better solution for implementing high speed, flexible, and scalable digital control systems.

5. Proposed solution:

The limitations of the traditional microcontroller-based vending machine controllers can be overcome by developing the system based on an FPGA. FPGA technology-based systems have better processing capabilities, parallel processing of operations, and greater flexibility in the design of digital systems. These characteristics make the FPGA technology more appropriate for developing complex control systems such as automated vending machines.

The following are the advantages that make the use of FPGA technology in the design of the vending machine controller system worthwhile:

FPGAs (Field Programmable Gate Arrays)

- **Parallel Processing:**

FPGA technology-based systems have the advantage of parallel processing, wherein multiple operations such as coin recognition, state changes, and output control can be performed concurrently.

- **Hardware Flexibility:**

FPGA technology-based systems have the advantage of hardware flexibility, wherein the system can be programmed and changed using hardware description languages such as Verilog HDL.

- **Precise Control:**

FPGA technology-based systems have better capabilities for precise control, wherein the system can provide accurate control of timing and signal generation, which is critical for the operation of digital control systems.

- **High Speed Operation:**

FPGA technology-based systems have the advantage of high-speed operation, wherein the system can operate at higher clock speeds compared to traditional microcontroller-based systems.

- **Design Learning and Implementation:**

FPGA-based systems require knowledge of designing digital logic circuits and the use of hardware description languages, such as Verilog, which gives a deeper insight into hardware-level design.

Thus, it may be concluded that though microcontrollers may be used to implement simpler automation-based projects, the flexibility, speed, and scalability offered by the use of FPGA-based systems make it a suitable choice to implement the controller of the vending machine.

Design assumptions:

The following assumptions are considered while designing the FPGA-based vending machine controller:

- The vending machine is assumed to be a two-product vending machine.
- The system is assumed to accept two types of coins with a denomination of 5rs and 10rs.
- The prices of the products are assumed to be 15rs for Product-1 and 20rs for Product-2.

- The coins are assumed to be inserted by switches on the FPGA development board.
- The output signals for the dispensing of products and return of change are assumed to be indicated by LEDs on the FPGA development board.
- The vending machine is assumed to be a finite state machine (FSM) with each state indicating the amount of money inserted.

6. State Diagram

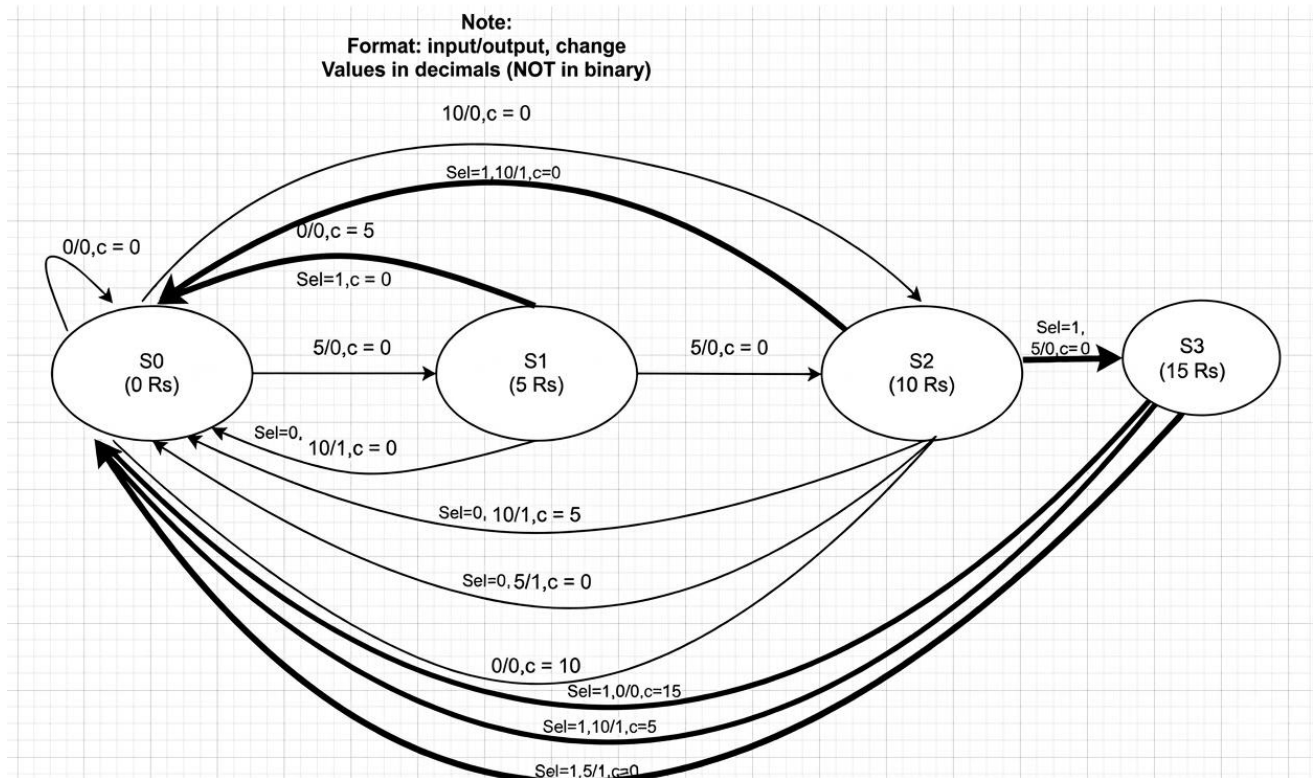


Fig.1. State diagram

7. Algorithm:

1. Initialization
 - Declare the module Vending_Machine_Controller.
 - Define input ports: clk, rst, in[1:0].
 - Define output ports: out, change[1:0].
 - Declare internal registers for current state (c_state) and next state (n_state).
2. Reset Condition
 - When reset signal is active, the vending machine returns to the initial state S0.
 - In this state the inserted amount is 0 Rs and outputs remain inactive.
3. Input Detection
 - The system continuously checks the input in[1:0] representing inserted coins.
 - 01 represents 5rs coin.
 - 10 represents 10rs coin.
4. State Transition
 - Based on the input coin value, the system transitions between states:
 - S0 to S1 when 5rs is inserted.

- S0 to S2 when 10rs is inserted.
 - S1 to S2 when another 5rs is inserted.
 - S2 to S3 when another 5rs is inserted.
5. Product Dispensing
 - When the required amount is reached (10rs), the controller activates the out signal to dispense the product.
 6. Change Return
 - If the inserted amount exceeds the required value, the controller activates the change output to return the extra amount.
 7. Return to Initial State
 - After dispensing the product and returning any change, the system resets back to state S0 and waits for the next input.

8. Flowchart

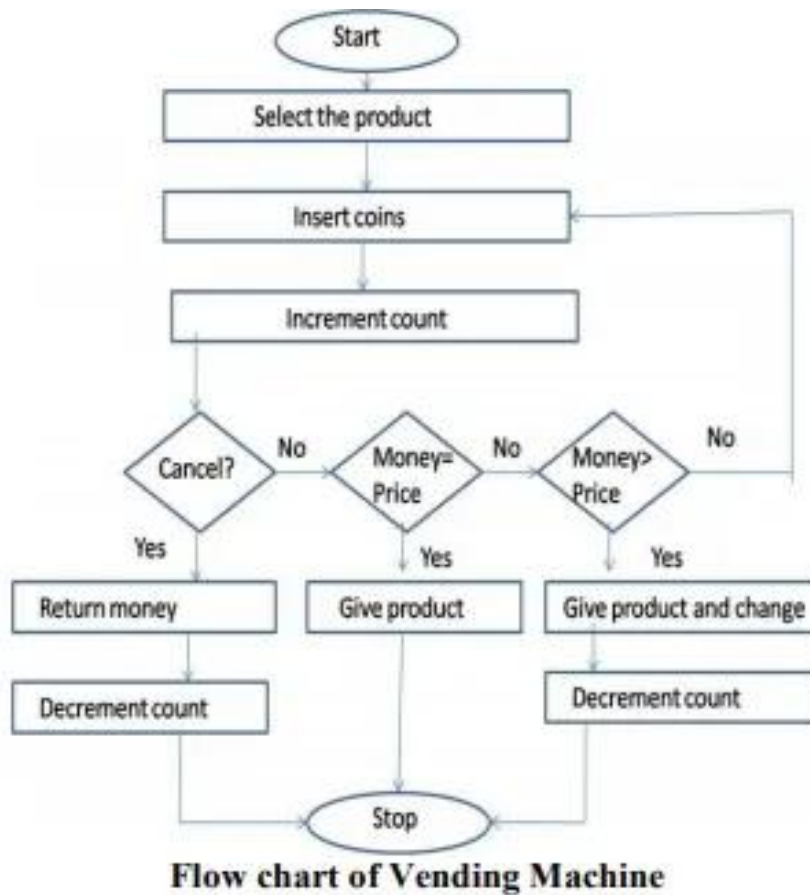


Fig.2. Flow diagram

9. RTL and TTL views:

1.1. RTL:

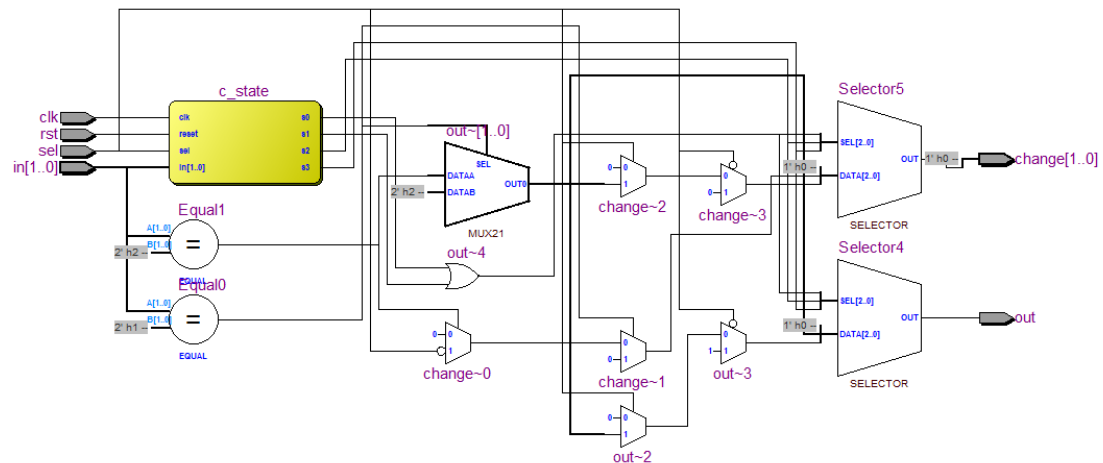


Fig.3. RTL

1.2. TTL:

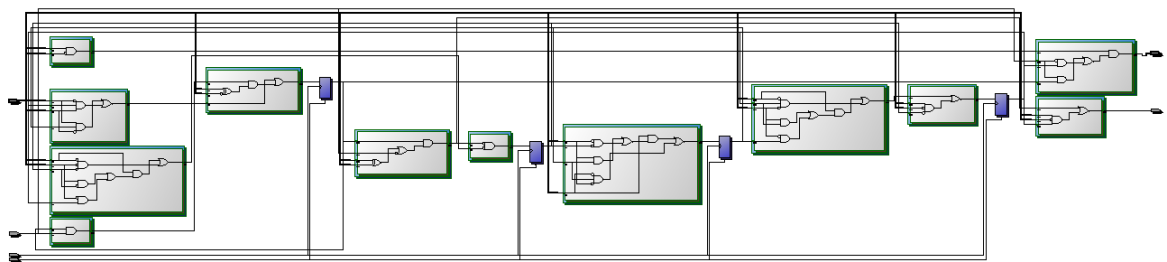


Fig.4. TTL

10. Simulation and result

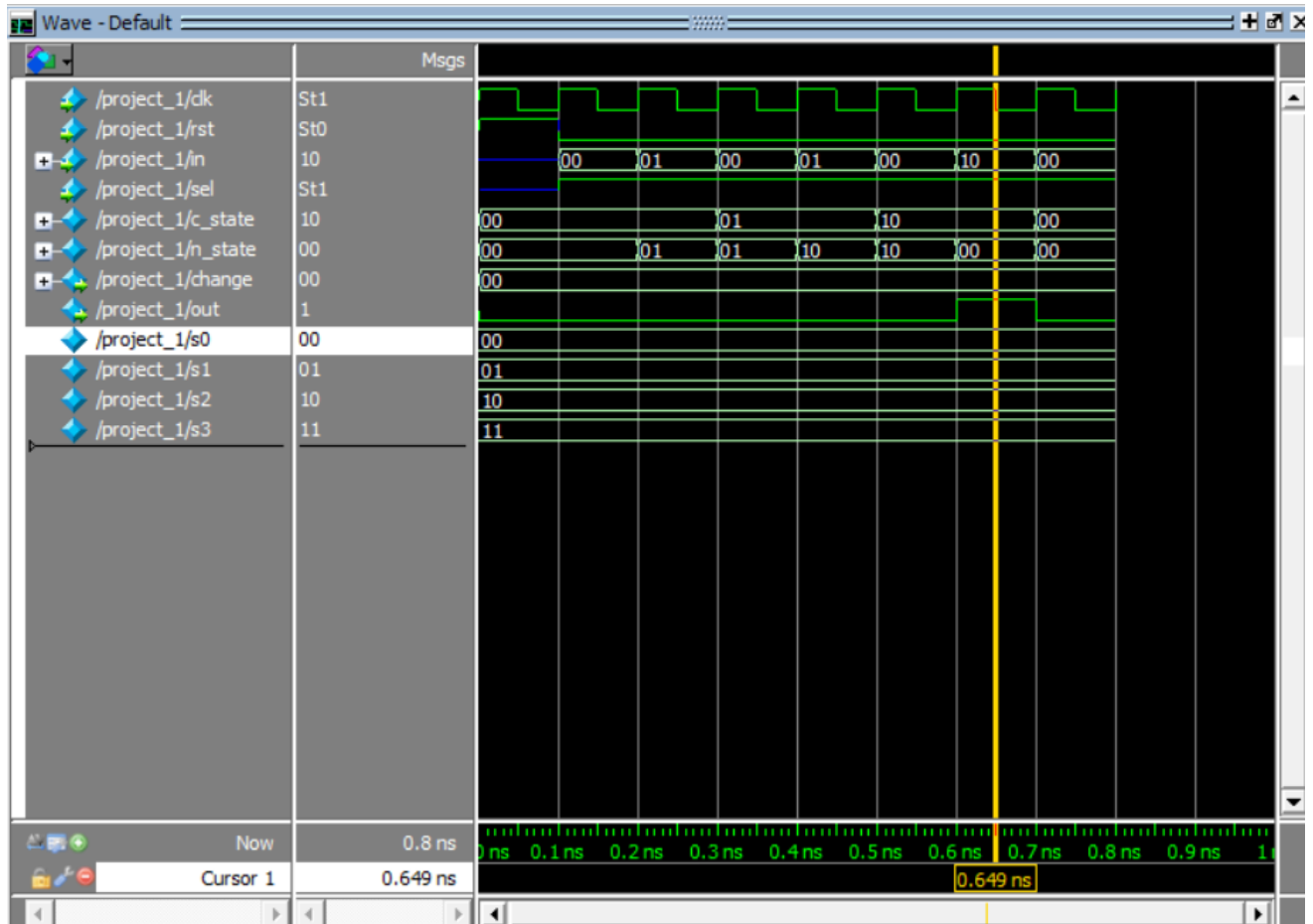


Fig 5. Modelsim Simulation Waveform

Result:

- The simulation showcases the operation of vending machine controller where the user entered the case of 20Rs product and initially the system is in reset therefore the state is S0 representing 0rs in the machine and then reset is set to 0 with a clock pulse of 100ps time period.
- The user initially inserts a 5Rs coin to the vending machine which transitions from S0(0Rs) to S1(5Rs) followed with another 5Rs coin hence, the total money stored in the vending machine is 10rs which transition the state S1(5Rs) to S2(10Rs).
- Since the selected product requires **20Rs** an additional **10Rs coin** is inserted. This increases the total credit to **20Rs**, satisfying the required amount for the selected product. Hence the product is dispensed with no change and we get the output as 1.

11. Conclusion:

Through the completion of this assignment, a better knowledge of Verilog HDL programming and the design of a digital system based on FPGAs was gained. The design and implementation of the controller for the vending machine helped to understand the concepts of finite state machines, simulation with the help of ModelSim, and synthesis with the help of Quartus.

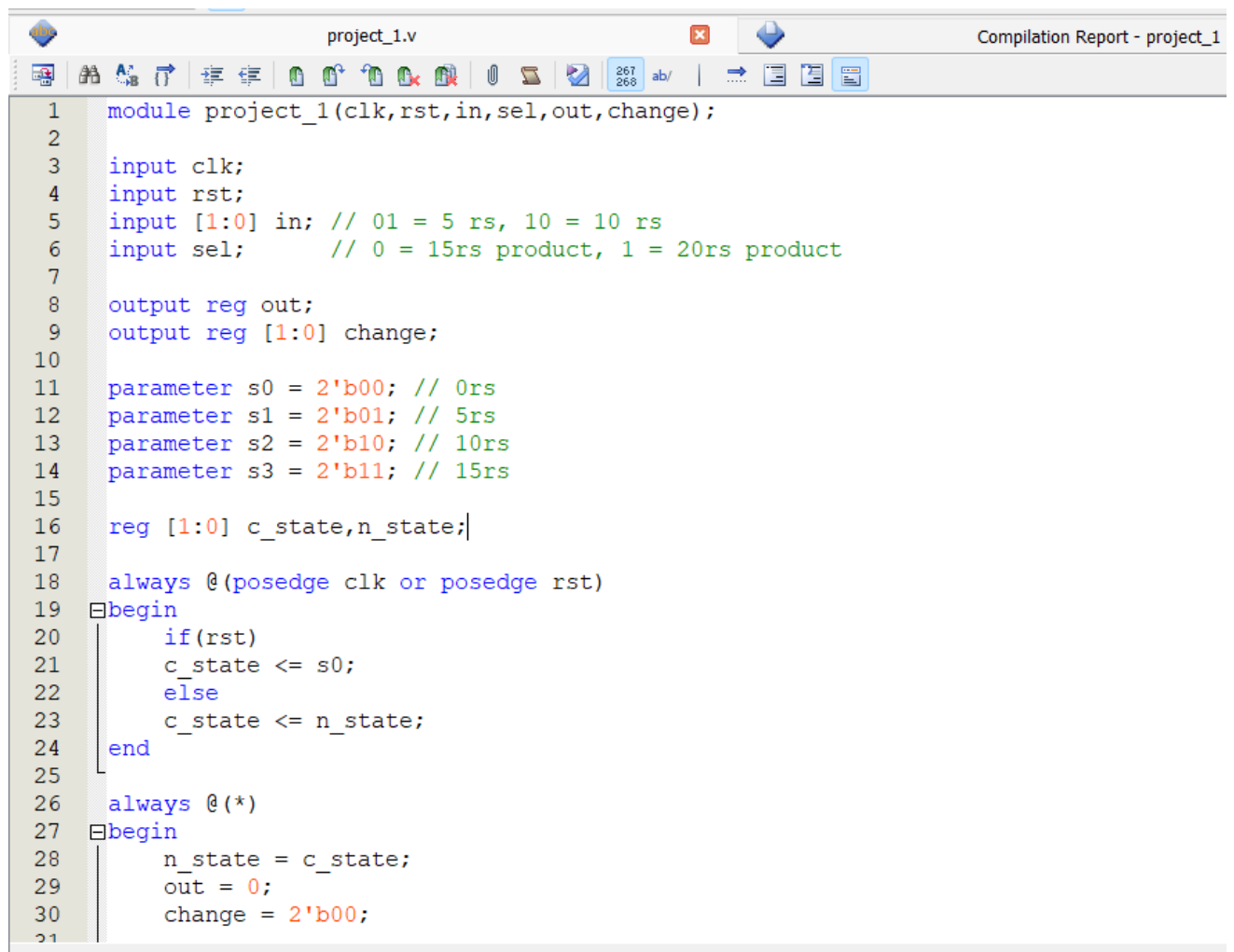
This project helped to gain hands-on experience with the implementation and testing of a digital system with the help of switches as inputs and LEDs as outputs. Overall, the assignment helped to gain knowledge of FPGAs.

12. References:

- [1] Samir Palnitkar, Verilog® HDL: A Guide to Digital Design and Synthesis, Prentice Hall PTR
- [2] Research Paper: (1) <https://medium.com/@rebsfalcao15/design-of-vending-machine-using-verilog-hdl-42cede6cf8bb>
(2) <https://www.scribd.com/presentation/422736906/Ppt1>
- [3] Reference video: <https://youtu.be/6NHmwnwmiaA?si=PcHqIUG84DmSHar->

13. Appendix:

13.1. Verilog HDL Code:



```
1 module project_1(clk,rst,in,sel,out,change);
2
3     input clk;
4     input rst;
5     input [1:0] in; // 01 = 5 rs, 10 = 10 rs
6     input sel;      // 0 = 15rs product, 1 = 20rs product
7
8     output reg out;
9     output reg [1:0] change;
10
11     parameter s0 = 2'b00; // 0rs
12     parameter s1 = 2'b01; // 5rs
13     parameter s2 = 2'b10; // 10rs
14     parameter s3 = 2'b11; // 15rs
15
16     reg [1:0] c_state,n_state;
17
18     always @(posedge clk or posedge rst)
19     begin
20         if(rst)
21             c_state <= s0;
22         else
23             c_state <= n_state;
24     end
25
26     always @(*)
27     begin
28         n_state = c_state;
29         out = 0;
30         change = 2'b00;
31     end
endmodule
```

```

28     n_state = c_state;
29     out = 0;
30     change = 2'b00;
31
32     case(c_state)
33     s0: // ₹0
34     begin
35         if(in == 2'b01)
36             n_state = s1;
37         else if(in == 2'b10)
38             n_state = s2;
39         end
40     s1: // ₹5
41     begin
42         if(in == 2'b01)
43             n_state = s2;
44         else if(in == 2'b10)
45             n_state = s3;
46         end
47     s2: // ₹10
48     begin
49         if(in == 2'b01)
50             n_state = s3;
51         else if(in == 2'b10)
52             begin
53                 if(sel == 0) // product 15
54                     begin
55                         ...
56                     end
57                 ...
58             end
59         end
60     end

```

```

55 | begin
56 |     if(sel == 0) // product 15
57 |     begin
58 |         out = 1;
59 |         change = 2'b01;
60 |         n_state = s0;
61 |     end
62 |     else // product 20
63 |     begin
64 |         out = 1;
65 |         n_state = s0;
66 |     end
67 | end
68 | end
69 |
70 | s3: // ₹15
71 | begin
72 |     if(sel == 0) // product 15
73 |     begin
74 |         out = 1;
75 |         n_state = s0;
76 |     end
77 |     else if(sel == 1) // need 5 more
78 |     begin
79 |         if(in == 2'b01)
80 |         begin
81 |             out = 1;
82 |             n_state = s0;
83 |         end
84 |         else if(in == 2'b10)

```

```

84 |         else if(in == 2'b10)
85 |         begin
86 |             out = 1;
87 |             change = 2'b01;
88 |             n_state = s0;
89 |         end
90 |     end
91 | end
92 |
93 | default:
94 | begin
95 |     n_state = s0;
96 |     out = 0;
97 |     change = 2'b00;
98 | end
99 |
100 | endcase
101 | end
102 |
103 | endmodule

```

THANK YOU